

Omega-3 Fatty Acids and incident Type 2 Diabetes: A Systematic Review and Meta-Analysis

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The relationship between omega-3 polyunsaturated fatty acids (n-3 PUFA) from seafood (eicosapentaenoic acid, EPA; docosahexaenoic acid, DHA) or plant (alpha-linolenic acid, ALA) sources and risk of type 2 diabetes mellitus (DM) remains unclear. We systematically searched multiple literature databases through June 2011 to identify prospective studies examining relations of dietary n-3 PUFA, dietary fish and/or seafood, and circulating n-3 PUFA biomarkers with incidence of DM. Data were independently extracted in duplicate by 2 investigators, including multivariate-adjusted relative risk (RR) estimates and corresponding 95% CIs. Generalized least-squares trend estimation was used to assess dose-response relationships, with pooled summary estimates calculated by both fixed-effect and random-effect models. From 288 identified abstracts, 16 studies met inclusion criteria, including 18 separate cohorts comprising 540,184 individuals and 25,670 cases of incident DM. Consumption of fish and/or seafood was not significantly associated with DM (n=13 studies; RR per 100g/d=1.12, 95% CI=0.94, 1.34); nor were consumption of EPA+DHA (n=16 cohorts; RR per 250mg/d=1.04, 95% CI=0.97, 1.10) or circulating levels of EPA+DHA biomarkers (n=5 cohorts; RR per 3% of total fatty acids=0.94, 95% CI=0.75, 1.17). Both dietary ALA (n=7 studies; RR per 0.5g/d=0.93, 95% CI=0.83, 1.04) and circulating ALA biomarker levels (n=6 studies; RR per 0.1% of total fatty acid=0.90, 95% CI=0.80, 1.00, P=0.06) were associated with non-significant trend towards lower risk of DM. Substantial heterogeneity (I²–80%) was observed among studies of fish/seafood or EPA+DHA and DM; moderate heterogeneity (<55%) was seen for dietary and biomarker ALA and DM. In unadjusted meta-regressions, study location (Asia vs. North America/Europe), mean BMI, and duration of follow-up each modified the association between fish/seafood and EPA+DHA consumption and DM risk (P-Interaction ≤ 0.02 each). We had limited statistical power to determine the independent effect of these sources of heterogeneity due to their high collinearity. The overall pooled findings do not support either major harms or benefits of fish/seafood or EPA+DHA on development of DM, and suggest that ALA may be associated with modestly lower risk. Reasons for potential heterogeneity of effects, which could include true biologic heterogeneity, publication bias, or chance, deserve further investigation.